

IS6620: LARGE LANGUAGE MODEL WITH PROMPT ENGINEERING FOR BUSINESS

Effective Term

Semester B 2025/26

Part I Course Overview

Course Title

Large Language Model with Prompt Engineering for Business

Subject Code

IS - Information Systems

Course Number

6620

Academic Unit

Information Systems (IS)

College/School

College of Business (CB)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

This Master-level course in the College of Business focuses on the utilization of large language models and prompt engineering techniques for enhancing business applications. Students will explore the capabilities of state-of-the-art language models, such as ChatGPT, and learn how to effectively tailor prompts to generate desired outputs. The course will cover topics including natural language processing, text generation, sentiment analysis, and language model fine-tuning. Through a combination of theoretical lectures, hands-on exercises, and real-world case studies, students will develop the skills to leverage large language models for various business use cases, such as customer service, content creation, and market research.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Design the principles and fundamentals of large language models.	10			
2	Analyze the concept of prompt engineering and its importance in achieving desired language model outputs.	20	x	x	
3	Gain proficiency in utilizing large language models for text generation and natural language processing tasks.	20	x	x	
4	Apply skills in fine-tuning language models for specific business applications.	20	x	x	
5	Apply large language models to solve real-world business challenges, such as customer sentiment analysis and content generation.	10	x	x	x
6	Analyze ethical considerations and potential biases associated with large language models in business applications.	10	x	x	
7	Relate critical thinking and problem-solving abilities through hands-on projects and case studies.	10	x	x	

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

	LTAs	Brief Description	CILO No.	Hours/week (if applicable)
1	LTA1: Seminars:	Students will learn the concepts, frameworks and strategies of large language models, its applications in business, and achieving desired language model outputs.	1, 2, 3, 4, 5, 6, 7	Seminar : 39 hours
2	LTA2: Case Studies:	Students will discuss and analyze the effective use of large language models (e.g., ChatGPT) to solve real-world business challenges.	1, 2, 3, 5	
3	LTA3: Hands-on Exercises:	Students will develop the practical skills of fine-tuning language models for specific business applications, utilize large language models for text generation and natural language processing tasks, and analyze ethical considerations and potential biases associated with large language models in business applications	2, 3, 4, 6	
4	LTA4: Project	Students would have to complete a group project to demonstrate the ability on effectively use large language model with prompt engineering for real-world business challenges.	1, 2, 3, 4, 5, 6, 7	

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	AT1. Participation and Exercises Each exercise consists of impromptu quizzes, paired/ small group discussions, role-plays, self reflection, or student presentations to assess students' understanding of the chosen topics and their abilities to apply their skills.	1, 2, 3, 4, 5, 6, 7	10	-	Yes
2	AT2. Group Project A group project, which includes a project report and/ or a presentation, will be assigned to let students apply large language model and prompt engineering concepts and techniques to plan, develop, and/ or evaluate the business solution for some real-world business challenge.	1, 2, 3, 4, 5, 6, 7	40	-	Yes

Continuous Assessment (%)

50

Examination (%)

50

Examination Duration (Hours)

2

Additional Information for ATs

AT3: Final Examination

A written examination is developed to assess student's competence level of the topics taught.

Note: Students must pass BOTH coursework and examination in order to get an overall pass in this course.

Assessment Rubrics (AR)

Assessment Task

AT1. Participation and Exercises (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to accurately describe all key concepts and principles of large language models;

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1. Participation and Exercises (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to accurately describe the concept of prompt engineering and its importance in achieving desired language model outputs;

Excellent

(A+, A, A-) High

Good

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Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1. Participation and Exercises (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to creatively and effectively use large language models for text generation and natural language processing tasks;

Excellent

(A+, A, A-) High

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(C+, C, C-) Moderate

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(D) Basic

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(F) Not even reaching marginal levels

Assessment Task

AT1. Participation and Exercises (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to creatively and effectively fine-tuning language models for specific business applications;

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1. Participation and Exercises (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to creatively and effectively apply large language models to solve real-world business challenges;

Excellent

(A+, A, A-) High

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Assessment Task

AT1. Participation and Exercises (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to accurately analyze ethical considerations and potential biases associated with large language models in business applications;

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

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Assessment Task

AT1. Participation and Exercises (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability on critical thinking and problem-solving

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT2. Group Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to accurately describe all key concepts and principles of large language models;

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

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AT2. Group Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

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Ability to accurately describe the concept of prompt engineering and its importance in achieving desired language model outputs;

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(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

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Failure

(F) Not even reaching marginal levels

Assessment Task

AT3. Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to accurately describe the concept of prompt engineering and its importance in achieving desired language model outputs;

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Failure

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Assessment Task

AT3. Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to creatively and effectively fine-tuning language models for specific business applications;

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT3. Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to creatively and effectively apply large language models to solve real-world business challenges;

Excellent

(A+, A, A-) High

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(B+, B, B-) Significant

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(C+, C, C-) Moderate

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Failure

(F) Not even reaching marginal levels

Assessment Task

AT3. Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to accurately analyze ethical considerations and potential biases associated with large language models in business applications

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(B+, B, B-) Significant

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(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1. Participation and Exercises (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to accurately describe all key concepts and principles of large language models;

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

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(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1. Participation and Exercises (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to accurately describe the concept of prompt engineering and its importance in achieving desired language model outputs;

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1. Participation and Exercises (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Capability to creatively and effectively use large language models for text generation and natural language processing tasks;

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1. Participation and Exercises (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Capability to creatively and effectively fine-tuning language models for specific business applications;

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1. Participation and Exercises (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Capability to creatively and effectively apply large language models to solve real-world business challenges;

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1. Participation and Exercises (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to accurately analyze ethical considerations and potential biases associated with large language models in business applications;

Excellent

(A+, A, A-) High

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(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1. Participation and Exercises (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability on critical thinking and problem-solving

Excellent

(A+, A, A-) High

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(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT2. Group Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to accurately describe all key concepts and principles of large language models;

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

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(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT2. Group Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to accurately describe the concept of prompt engineering and its importance in achieving desired language model outputs;

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Assessment Task

AT2. Group Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Capability to creatively and effectively fine-tuning language models for specific business applications;

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AT2. Group Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

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AT2. Group Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

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Ability to accurately analyze ethical considerations and potential biases associated with large language models in business applications;

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AT2. Group Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

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AT3. Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

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Assessment Task

AT3. Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

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AT3. Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

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Capability to creatively and effectively fine-tuning language models for specific business applications;

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Criterion

Capability to creatively and effectively apply large language models to solve real-world business challenges;

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Assessment Task

AT3. Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to accurately analyze ethical considerations and potential biases associated with large language models in business applications;

Excellent

(A+, A, A-) High

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(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Part III Other Information

Keyword Syllabus

Introduction to large Language Models

- Overview of large language models and their significance in business
- Deep dive into ChatGPT and other prominent language models
- Ethical considerations and biases in large language models

Prompt Engineering for Desired Outputs

- Importance of prompt engineering in achieving desired results
- Techniques for crafting effective prompts and instructions
- Strategies for refining and iterating prompts

Natural Language Processing with Large Language Models

- Sentiment analysis using language models
- Named entity recognition and information extraction
- Text classification and clustering

Text Generation and Content Creation

- Text completion and expansion with large language models
- Generating product descriptions and marketing copy
- Automated content creation for social media and blogs

Language Model Fine-Tuning for Business Applications

- Overview of fine-tuning techniques
- Domain-specific language model customization
- Fine-tuning for sentiment analysis and other business tasks

Business Use Cases for Large Language Models

- Customer service and chatbot applications
- Market research and consumer insights
- Content moderation and filtering

Case Studies and Real-World Applications

- Analysis and discussion of successful business applications of large language models
- Hands-on projects applying large language models to business challenges

- Ethical considerations and potential risks in business applications
- Future Directions and Emerging Trends
- Exploration of advancements in large language models
 - Implications of emerging models and technologies on business applications
 - Opportunities and challenges for large language models in the future

Reading List

Compulsory Readings

Title	
1	Lecture notes in class

Additional Readings

Title	
1	Nil